



PROJECT ON

LED Chaser Circuit using 555 Timer IC + 4017 IC

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Course Title : Digital Electronics Sessional

Submitted To:

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Name Of the project:

LED Chaser circuit using 555 timer + 4017 IC

Introduction:

The LED Chaser Circuit is a popular electronics project that utilizes a 555 timer IC and a 4017 decade counter IC. This project creates a visual sequence where LEDs light up in a chasing pattern. The circuit is widely used in decoration, signaling, and learning applications due to its simplicity and effectiveness in demonstrating basic electronics concepts.

Project Objective:

Educational Purpose:

- To provide a practical demonstration of basic electronic components like the 555 timer and 4017 IC.
- To enhance understanding of how timing circuits and counters work in electronics.

Circuit Design and Implementation:

- To design and build a fully functional LED chaser circuit that illustrates the sequential switching concept.
- To use the circuit as an example for understanding the integration of analog (555 timer) and digital (4017 IC) electronics.

Hands-On Skills Development:

- To gain hands-on experience in assembling, testing, and troubleshooting electronic circuits on a breadboard.

Application-Oriented Learning:

- To explore the practical uses of LED chasers in real-world scenarios such as decorative lighting, visual indicators, and educational tools.

Adaptability and Customization:

- To encourage circuit modifications by varying LED numbers, adding features like speed control potentiometers, or integrating microcontroller-based enhancements.

Logic Functionality:

The circuit relies on the following functionalities:

- The 555 timer generates a continuous square wave.
- The 4017 IC acts as a decade counter, decoding the timer pulses into sequential outputs.
- LEDs connected to the outputs of the 4017 IC turn on in a pre-determined sequence, creating the chasing effect.

Truth Table and State Diagram:

Truth Table (Simplified for 4 LEDs):

Clock Pulse	Q0	Q1	Q2	Q3
1	1	0	0	0
2	0	1	0	0
3	0	0	1	0
4	0	0	0	1
Repeat	..	..	..	..

State Diagram:

State transitions occur sequentially as the clock pulses from the 555 timer advance the 4017 IC outputs.

Component Selection:

555 Timer IC:

- Functions as a clock pulse generator.
- Operates in astable mode to produce a continuous square wave signal.

- Provides flexibility in controlling the frequency of pulses through external resistors and capacitors.

4017 IC (Decade Counter):

- Acts as a sequential counter to drive the LEDs in a chasing pattern.
- Has 10 outputs (Q0 to Q9), with only one active at a time.
- Includes a reset pin to restart the sequence from the beginning.

LEDs (Light Emitting Diodes):

- Visual indicators of the sequential outputs from the 4017 IC.
- Different colors can be used for aesthetic appeal.
- Require current-limiting resistors to prevent damage.

Resistors:

- Current-Limiting Resistors: Connected in series with LEDs to protect them from excessive current.
- Timing Resistors: Used with the 555 timer IC to set the frequency of clock pulses.
- Typical values range from a few ohms (for LEDs) to kilo-ohms (for timing).

Capacitors:

- Timing Capacitor: Determines the pulse width and frequency of the 555 timer IC.
- Decoupling Capacitors: Placed near the ICs to filter noise and stabilize the power supply.

Potentiometer (Optional):

- Allows dynamic adjustment of the pulse frequency, offering real-time speed control of the LED chaser.

Power Supply:

- Provides the required voltage (typically 5V or 9V DC) to the circuit.
- A battery or DC adapter can be used, ensuring compatibility with the ICs and LEDs.

Connecting Wires and Breadboard:

- Wires are used to establish connections between components.
- A breadboard facilitates quick assembly and testing of the circuit.

Timing Constraints:

- The timing capacitor and resistor values for the 555 timer must be chosen carefully to set the desired clock pulse frequency.
- The LED on-time and off-time are directly proportional to the clock frequency.

Power Supply and Voltage Levels:

- The circuit operates on a regulated 5V or 9V DC power supply.
- Ensure that the 555 timer and 4017 IC voltage levels are compatible.

Schematic Design:

The Design is already shown the simulation and circuit diagram. All function and table showed in the report.

Simulation:

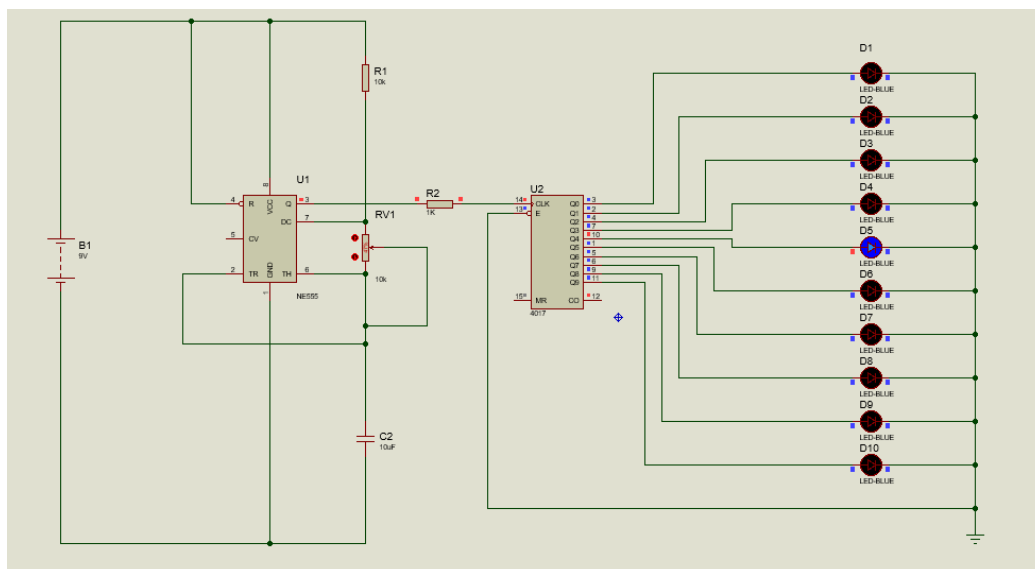


Fig: Simulation Diagram for LED Chaser Circuit

Prototyping Platform:

We completed the connection of the circuit in breadboard.

Programming:

Programming is not applicable in the project.

Documentation:

The design, implementation, the testing process, the truth table, state diagram all is show that in report.

Cicuit Diagram:

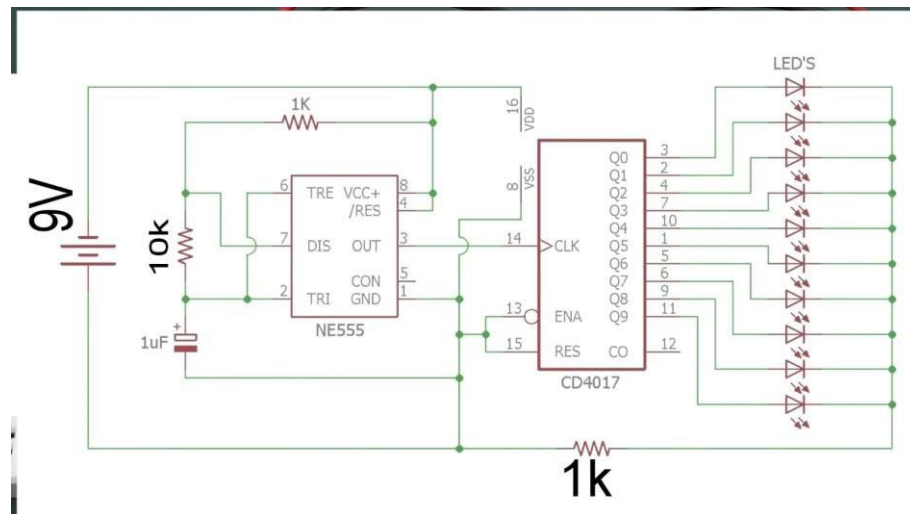


Fig: Circuit diagram for LED Chaser

Working Principle:

The LED chaser circuit works by combining the clock-generating capabilities of the 555 timer IC with the sequential output of the 4017 decade counter IC. Here is a detailed breakdown of the working principle:

1. Clock Pulse Generation (555 Timer IC):

- The 555 timer IC is configured in astable mode, where it generates a continuous series of square wave pulses. These pulses act as a clock signal for the 4017 IC.
- The frequency of the clock pulses depends on the values of the timing resistor (R) and capacitor (C) connected to the 555 timer.

- The duty cycle of the clock pulse ensures that each LED stays on for a specific amount of time before the next LED is activated.

2. Sequential Switching (4017 IC):

- The 4017 IC is a decade counter that advances its output by one step for every clock pulse received at its clock input.
- The IC has ten output pins (Q0 to Q9), but only one pin is active (HIGH) at any given time.
- When the clock pulse is received, the active HIGH output shifts sequentially from one pin to the next, creating the "chasing" effect.

3. LED Illumination:

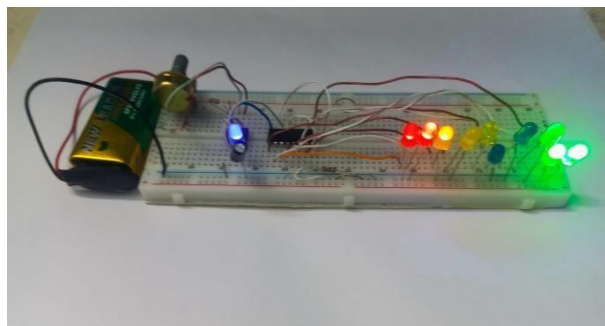
- LEDs are connected to the output pins of the 4017 IC through current-limiting resistors.
- As each output pin becomes active, the corresponding LED lights up, creating the visual effect of motion.

4. Reset Functionality:

- The 4017 IC is designed to reset back to the first output (Q0) once it reaches the last output (Q9) or a predefined count, depending on the wiring.
- This ensures a continuous looping sequence of LED illumination.

5. Speed Control:

By adjusting the resistor and capacitor values of the 555 timer, the frequency of the clock pulses can be increased or decreased, directly controlling the speed of the LED chaser.



Picture of the Project

Conclusion:

The LED Chaser Circuit effectively combines analog and digital electronics to create an engaging visual output. Through the integration of the 555 timer IC and 4017 decade counter IC, the project demonstrates fundamental principles of timing, counting, and sequential logic. It is an excellent project for students and hobbyists to learn practical electronics, offering insights into circuit design and functionality. This circuit is also highly adaptable, with room for customization, such as varying the number of LEDs, changing the speed control mechanism, or integrating additional features like microcontroller-based controls for more complex patterns. The simplicity and versatility of the LED Chaser Circuit make it a staple project in electronics education and a foundation for advanced explorations in sequential circuits and visual effects design.

By successfully completing this project, learners develop essential skills in circuit assembly, testing, troubleshooting, and design optimization. The project serves not only as an educational tool but also as a gateway to exploring practical applications in decorative lighting, visual indicators, and beyond.

Cost Calculation:

1. Bread Board.....	120TK
2. NE555 Timer IC (1p)	20TK
3. CD4017 IC (1p).....	15TK
4. LED (10p)	20TK
5. Resistor(1k,10k)	10TK
6. Potentiometer.....	15TK
7. Capacitors(10 μ F).....	5TK
8. 9V Battery	60Tk
9. Connecting Wires	10TK

Total.....275TK

Contribution of the Group Member:

ET-233053 (Md. Nazmul Hossan Talukder Fahim)

Role: Lead the project, report writing and designing, buying equipment.

ET-233057 (Mahadin Jakir)

Role: Designing and testing specialist, circuit implementing and developing.

ET-233045 (Syed Shafin Taosif)

Role: Data Collection and technical expert.

Overall, we worked and implement the circuit, practicing simulating in proteus software. We finished the project successfully.